

Slicing PBFT Consensus Algorithm Based on VRF

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Abstract—Blockchain technology has become a key support for optimizing Consortium Blockchain processes and improving information sharing efficiency. However, existing consortium blockchain consensus algorithms often exhibit declining efficiency and insufficient stability when facing large-scale networks and frequent membership changes, constraining their application in complex Consortium Blockchain scenarios. To address these issues, this paper proposes a Practical Byzantine Fault Tolerance based on Verifiable Random Function (VRSPBFT). VRSPBFT uses the node's VRF proof value as the election basis, ensuring that more credible and contributive nodes are more likely to become leaders, avoiding malicious abuse. The model also designs a VRF-based instant node joining mechanism, allowing new nodes to quickly join the consensus process without waiting, effectively alleviating the consensus interruption problem caused by frequent changes and improving overall efficiency and continuity. The VRSPBFT model not only provides new technical support for building secure and reliable blockchain-based Consortium Blockchain systems, but also offers valuable references for further optimizing consortium blockchain consensus algorithms. This helps promote the application of blockchain in complex Consortium Blockchain scenarios and facilitate the transparency and efficiency of Consortium Blockchain management.

Index Terms—Blockchain, VRF, PBFT, RSPBFT

I. INTRODUCTION

Blockchain technology has been gradually applied to Consortium Blockchain management, becoming a key support for optimizing processes and improving information sharing. Blockchain can effectively reduce transmission time and enhance trust among participants, promoting collaboration.

However, existing consortium blockchain consensus algorithms often exhibit declining efficiency and insufficient stability

when facing large-scale networks and frequent membership changes, constraining their application in complex Consortium Blockchain scenarios. The PBFT algorithm, as a typical mechanism, suffers from performance bottlenecks with large-scale participation, particularly in the primary node election and dynamic node joining.

Existing solutions have difficulty ensuring fairness in primary node election, with potential risks of abuse from malicious nodes. Regarding dynamic node joining, the immediate addition of new nodes may cause consensus interruptions, reducing efficiency and continuity. These problems severely limit PBFT's applicability in large-scale Consortium Blockchain scenarios.

To address the issues in existing PBFT-based algorithms, scholars have proposed the RSPBFT scheme, which improved consensus efficiency in large-scale Consortium Blockchain networks to some extent. However, RSPBFT still needs to further enhance the fairness of primary node election and optimize the impact of dynamic node joining.

To this end, this paper proposes the VRSPBFT model, which introduces VRF technology to design a fair primary node election protocol. VRSPBFT uses the node's VRF proof value as the election basis, ensuring more credible and contributive nodes are more likely to become leaders, avoiding malicious abuse. The verifiability of VRF also enhances the trust relationship among Consortium Blockchain participants and the reliability of consensus decisions.

Regarding dynamic node joining, VRSPBFT designs a VRF-based instant mechanism, allowing new nodes to quickly

complete authentication and verification without waiting, effectively reducing the interruptions caused by frequent node changes. This significantly improves the consensus efficiency and continuity of the entire Consortium Blockchain system.

The VRSPBFT model fully utilizes VRF technology to achieve fairness in primary node election and smoothness in dynamic node joining, while maintaining high-efficiency consensus. This provides a new solution for building more secure and reliable blockchain-based Consortium Blockchain systems, and offers valuable references for further optimizing consortium blockchain consensus algorithms, contributing to the advancement of blockchain technology in complex Consortium Blockchain scenarios.

The main contributions of this paper can be summarized as follows:

- VRSPBFT introduces VRF technology to design a fair primary node election protocol. By using the node's VRF proof as the election basis, VRSPBFT ensures that more credible and contributive nodes are more likely to be elected as primary nodes, avoiding the risks of malicious abuse. The verifiability of VRF proofs further enhances the trust of Consortium Blockchain participants in the consensus process, supporting the development of blockchain-based Consortium Blockchain trust.
- VRSPBFT also designs an instant node joining mechanism, allowing new nodes to quickly authenticate and join the consensus without waiting, effectively alleviating the consensus interruption caused by frequent node changes. This not only solves the PBFT performance bottleneck in large-scale Consortium Blockchain networks, but also provides a valuable reference for optimizing consortium blockchain consensus algorithms.

The remainder of this paper is organized as follows: Section 2 provides the fundamental knowledge on PBFT and VRF, including the roles and algorithm process of PBFT, as well as the properties and construction of VRF. Section 3 presents the detailed design and implementation of the VRSPBFT (VRF-based Slice PBFT) consensus scheme. Section 4 discusses the security analysis and performance evaluation of the proposed approach. Finally, Section 5 concludes the paper and outlines future research directions.

II. RELATED KNOWLEDGE

Regarding the limitations of the most widely used algorithms like PBFT in terms of scalability, efficiency, and adaptation to modern network conditions, the research focus in the Consortium Blockchain and IoT (Internet of Things) domains has shifted towards the deep optimization of their core processes and the revolutionary redesign of the underlying architecture [1]. Such research aims to overcome the performance bottlenecks of the original algorithms under large-scale node deployment, low-latency communication requirements, and dynamic network conditions, while also strengthening the security defense against various Byzantine fault modes.

Against this backdrop, a series of advanced Byzantine-fault-tolerant (BFT) consensus algorithms have emerged, which

not only effectively compensate for the deficiencies of the PBFT algorithm at the theoretical level, but also demonstrate remarkable performance improvements and enhanced adaptability in practice, strongly supporting the efficient, secure, and reliable decentralized consensus of blockchain technology in complex network environments. These advancements have not only enriched the technical arsenal of distributed consensus theory but also provided a solid theoretical foundation and engineering practice for the construction of more robust and widely applicable next-generation blockchain infrastructure.

To optimize the performance of permissioned blockchains, the tPBFT algorithm [2] introduces a trust-based fairness scoring mechanism, allowing for the dynamic adjustment of consensus nodes and simplifying the consensus process. Compared to the PBFT algorithm, tPBFT demonstrates superior performance in terms of consensus efficiency and node scalability. The HotStuff algorithm improves the PBFT network topology to a star-shaped network and adopts threshold signature technology, where the leader node collects signatures centrally, replacing the traditional many-to-many broadcast communication, while also employing parallel pipelined transaction processing. Although the HotStuff algorithm significantly improves consensus efficiency, the increased communication burden on the leader node may lead to network congestion. The DPNPBFT algorithm [3] employs dual leader nodes to prevent excessive centralization of a single leader, achieving good system security. The AnonymousFox consensus algorithm [4] designs an anonymous leader node ordering algorithm to hide the leader's identity and reduces node communication overhead through one-to-many message communication, thereby improving the algorithm's throughput and system security.

The number of nodes participating in the PBFT consensus process has a significant impact on the communication complexity of the algorithm. Accordingly, many researchers are focusing on exploring methods to reduce the number of nodes involved in the consensus process. By decreasing the number of nodes, the network communication burden can be reduced, thereby improving the algorithm's efficiency and performance. Algorithms based on the "committee + PBFT" model [5] have been proposed, where a subset of nodes is selected from the system as a committee based on factors such as node credibility and tokens. These committee nodes represent the entire network in the consensus protocol, while other nodes are responsible for verification and synchronization, enabling rapid consensus within the network. To avoid excessive centralization of the selected nodes, the Algorand algorithm [6] uses VRF [7] to randomly select a subset of network nodes as the committee, ensuring the unpredictability of the committee members. Although the "committee + PBFT" mode has improved algorithm performance and scalability, the excessive power of the committee nodes violates the decentralization principle of blockchain.

The main problems with the PBFT algorithm are its high communication complexity and lack of scalability. Optimizing PBFT alone cannot meet the requirements of large-scale

applications. Therefore, researchers have proposed a series of new consensus frameworks, whether adopting horizontal sharding or vertical layering, whose core is to divide nodes into independent spaces.

Elastico [8] is the first Byzantine-fault-tolerant algorithm that uses sharding technology. It uses a PoW (Proof-of-Work) algorithm to competitively select committee nodes from the network and assigns these nodes to different groups. Each group uses the PBFT algorithm to process different transaction sets in parallel, thereby improving the system's performance through parallel processing.

Since BFT algorithms have limits on the number of malicious users, if users collude with malicious users and exceed this limit, consensus cannot be reached. To solve this problem, Jeon et al. [9] propose a Byzantine-fault-tolerant random mesh blockchain that uses opinion diversity. For specific blockchain application scenarios, some efficient network partitioning methods have also been proposed, which have performed well in practical applications.

Lin et al. [10] propose a group strategy based on link identification, whose supervisory mechanism is more suitable for the supervisory system of a consortium chain. Yoo et al. [11] propose a location-based sharding method, which uses an improved K-medoids clustering algorithm to cluster the consensus nodes and select the nodes with higher trust as the cluster centers.

However, optimization from the perspective of a flat structure is no longer sufficient, and more and more research is beginning to explore hierarchical structures. Feng et al. [12] propose a scalable dynamic multi-agent hierarchical SDMA-PBFA algorithm, in which the consensus model divides the nodes into multiple groups, each of which selects a node to act as a proxy to execute the PBFT algorithm, and the resulting consensus is then sent to the clients.

Li et al. [13] have made multiple optimizations based on PBFT, designing a hierarchical structure to accelerate the consensus process and achieving global consensus through local consensus. Furthermore, Li et al. [14] have proposed a scalable multi-layer consensus model based on the PBFT algorithm, which greatly reduces the communication complexity and provides valuable guidance for the design and performance analysis of hierarchical systems.

III. MODEL DESIGN

A. VRSPBFT Framework

The VRSPBFT consensus model is divided into three distinct regions: **Primary-Area**, **Bottom-Area**, and **Gray-Area**.

Primary-Area:

- Located at the top of the consensus model, it plays a crucial coordinating and decision-making role.
- The primary nodes in the Primary-Area are responsible for compiling and summarizing the ordering information from the Bottom-Area, generating new blocks, and uploading them to the blockchain, ensuring the continuous updating and integrity of the blockchain.

- The secondary nodes (backup nodes) in the Primary-Area are responsible for coordinating the consensus messages from the Bottom-Area and transmitting them to the primary nodes.

Bottom-Area:

- Located at the bottom of the consensus model, it is responsible for message collection and initial consensus.
- The primary nodes in the Bottom-Area are responsible for processing and confirming the consensus messages within the region, and transmitting the consensus messages to the upper layer.
- The primary nodes in the Bottom-Area also serve as the secondary nodes in the Primary-Area, connecting the two regions to ensure the smooth operation of the entire consensus model.

Gray-Area:

- It encompasses a large number of nodes that may be unable to participate in the primary consensus process due to performance, reputation, or other reasons.
- By isolating the nodes in the Gray-Area, it not only enhances the stability of the entire consensus scheme, but also effectively reduces the negative impact of Byzantine nodes.
- This design effectively isolates potential instability factors, improving the system's inclusiveness and robustness.

The VRSPBFT defines three key roles in its structure: Primary Nodes, Backup Nodes, and Gray Nodes.

Primary Nodes:

- In the Primary-Area, they are responsible for compiling and ordering the messages from the Bottom-Area, packaging them into blocks, and adding them to blockchain.
- In the Bottom-Area, they are responsible for uploading the consensus messages to the Primary-Area in order.
- Primary nodes in Bottom-Area also serve as secondary nodes in Primary-Area, connecting the two regions.

Backup Nodes:

- They are responsible for packaging and backing up the original information, and actively participating in the consensus process of their respective regions.
- When they detect abnormalities in the primary nodes, they trigger the primary node election protocol to ensure system stability.

Gray Nodes:

- They are nodes with low reputation, unstable behavior, or exhibiting Byzantine behavior.
- By separating malicious nodes from normal nodes, the negative impact of malicious nodes on the entire system is reduced, and the system stability is improved.

In summary, the VRSPBFT, through the collaboration of these three regions and three types of roles, constructs an efficient, reliable, and fault-tolerant consensus model, as shown in the fig.1.

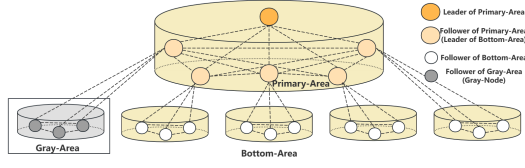


Fig. 1. VRSPBFT Framework

B. Dynamic node joining

Due to the randomness of node joining, the consensus process needs to be interrupted when a node requires joining the consensus network area, and then resumed at the interrupted point after the node has joined. Therefore, it is necessary to record the unique sequence number n of message m at the time of interruption. After completing the dynamic node joining mechanism, the consensus can then continue for the unique sequence number n of message m . This approach improves the efficiency of the consensus.

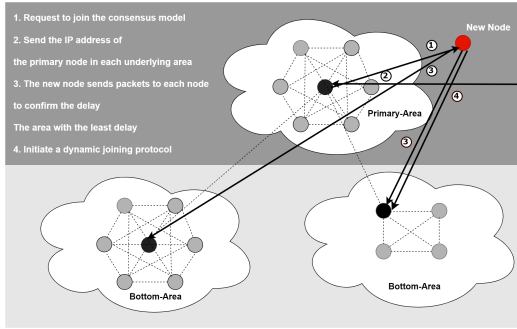


Fig. 2. Dynamic node joining

After being authorized, node i will also receive the IP addresses of the leader nodes of the lower-level regions. After receiving the IP addresses, i will choose a node region with low latency and less than the adjustable upper limit factor k of the number of nodes in the region. Then node i will send a $\langle join-req, c, mac, ip, pk \rangle_{Sig_i}$ message to the leader node of the selected region, where c represents the authorization certificate, mac represents the node's MAC address, ip represents the IP address of node i , and pk represents the public key of node i . The leader node will broadcast this message to the nodes in the region. The nodes in the region will verify the authorization certificate c of node i and record the message in their local logs.

The current leader node p in the region will then broadcast a $\langle \langle pre-join, c, mac, ip, pk, d, n, v, t, c \rangle_{Sig_p}, m \rangle$ message to all the replica nodes in the region, where m represents the current consensus message, d is the hash digest of message m , n is the unique sequence number assigned by the leader node to message m , v is the current view number, t is the unique sequence number assigned by the leader node to this sub-protocol message, and c is the sequence number of the latest stable checkpoint. After receiving the Pre-join message,

the replica nodes will compare it with the received Join-req message and verify the correctness of the message.

If the verification is successful, the new node j will broadcast a $\langle join, d, n, v, i, j, c \rangle_{Sig_j}$ message in the region. When node q receives $2f + 1$ different Join messages, it means the region nodes have reached consensus on node i 's entry, and node q will write node i 's information to its local NIT table and send a $\langle join-reply, d, n, v, i, j \rangle_{Sig_q}$ message to node i . After node i receives $f + 1$ Join-reply messages, it can join the consensus region.

The node dynamically joins protocol:

step 1: The new node will send its own *NodeInfo* to the leader node of the Primary-area.

step 2: The Primary-area leader node will verify the correctness of the *NodeInfo*. If the verification is successful, it will return an approval feedback message and the IP addresses of the leader nodes of the Bottom-areas.

step 3: The new node will send packets to the leader nodes of each Bottom-area, generate a node delay table PingList, and find the leader node of the Bottom-area with the minimum delay in the PingList.

step 4: The new node will send a $\langle join-req, c, mac, ip, pk \rangle_{Sig_i}$ message to the selected Bottom-area leader node.

step 5: The leader node will broadcast this message within the Bottom-area.

step 6: After receiving the message, the nodes in the Bottom-area will verify the new node's authorization certificate and record the message in their local logs.

step 7: The leader node will broadcast a $\langle \langle pre-join, c, mac, ip, pk, d, n, v, t, c \rangle_{Sig_p}, m \rangle$ message.

step 8: After verification by all the nodes, the new node will broadcast a $\langle join, d, n, v, i, j, c \rangle_{Sig_j}$ message in the Bottom-area. When $2f + 1$ different Join messages are received, it means the Bottom-area nodes have reached consensus on node i 's entry.

step 9: The node i 's information will be written into the local NIT table, and a $\langle join-reply, d, n, v, i, j \rangle_{Sig_q}$ message will be sent to node i .

step 10: After receiving $f + 1$ Join-reply messages, node i can join the consensus Bottom-area. The information of the new node's joining Bottom-area leader node will be output.

C. Improved Leader election

In the RSPBFT consensus algorithm environment, VRF is creatively applied to solve the randomness trust problem in the leader election process. Specifically, the unpredictable and provably fair random number generation mechanism provided by VRF can strongly support the transparency and fairness of the leader election in a decentralized environment, thereby helping to enhance the overall trustworthiness and balance of the RSPBFT blockchain system. This avoids the centralization advantages accumulation, i.e., the "rich getting richer" effect, caused by biases in random number generation. Additionally, the unpredictability of the leader node achieved through the presence of multiple highest-reputation nodes solves the single

point attack problem. By introducing VRF, RSPBFT can more robustly realize an equal and non-manipulable leader node selection process among the participating nodes. 3

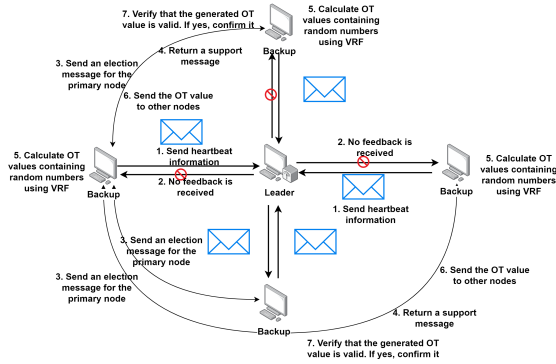


Fig. 3. Leader election

here are the steps for the improved VRF in RSPBFT:

Step 1: The participating nodes send a heartbeat message to the leader node, containing $\langle NIT, ln, nn_i, cn, cersco, \eta_k, cersco_\beta, v, \theta \rangle$.

Step 2: After waiting for a certain period of time without receiving a feedback message from the leader node, the nodes enter the pre-preparation stage.

Step 3: The nodes send a $\langle pre - leader - election, v, i, T \rangle_{sig_i}$ message to each node in the region, and wait for T time, where:

$$T = \begin{cases} \Delta T & , T < \Delta T \\ T & , T \geq \Delta T \end{cases} \quad (1)$$

Step 4: The parameter ΔT represents the network anomaly time coefficient. When T is greater than or equal to ΔT , this avoids the performance degradation and resource waste caused by frequent leader node elections due to common network issues like network fluctuations.

Step 5: After receiving the Pre-leader-election message, the other nodes verify whether the leader node has lost connection. If the disconnection is confirmed, they return a confirmation message.

Step 6: After receiving $2f + 1$ confirmation messages from the region, the participating nodes enter the leader node election stage. The participating nodes calculate a random number T'' within $(0, T]$ using VRF, and obtain:

$$OT_j = T + \frac{\varepsilon}{R_j^T} \quad (2)$$

Step 7: Where R_i^T represents the reputation evaluation score of node i , and ε is the leader node election time coefficient. $\frac{\varepsilon}{R_j^T}$ is used to calculate the waiting time for node i to send the leader node election request, ensuring that nodes with higher reputation have a higher probability of initiating the leader node election request first. The node will then broadcast a $\langle leader - election, v, i, OT_i \rangle_{sig_i}$ message in the current region at the time OT_i .

Step 8: Once the leader-election message is received, the nodes verify whether the OT time is normal. If it is abnormal, the message is discarded. If it is normal, the nodes will verify their local term information. If the term number in the message is less than their own recorded term number, the message will be immediately discarded. If the term number in the heartbeat message is greater than or equal to the node's own recorded term number, the node will accept the message.

Step 9: If the node has not yet returned a vote in the current term, it will return a confirmation message signed by its own private key. Once the node receives votes from more than half of the leader nodes, it is elected as the leader node of the region, and the term number will increase monotonically by 1.

The VRSPBFT has successfully solved the fairness issue of "the rich getting richer" and the security issue of single point attack that existed in the original RSPBFT due to its unique leader node election protocol under certain circumstances.

IV. EXPERIMENTAL AND ANALYSIS

This section presents experimental results that verify the actual effects of the improved RSPBFT algorithm. The results demonstrate the fairness and efficiency achieved by the proposed mechanisms in enhancing the positivity of commodity Consortium Blockchain applications. By optimizing the dynamic node joining and leader election mechanisms using RSPBFT, the issues of unfairness and inability of dynamic node joining in the Consortium Blockchain have been successfully resolved, providing strong support for its practical application feasibility.

A. Time complexity

Assume that the Primary-Area leader has m backups, and each backup acts as the leader for n nodes in its corresponding Bottom-Area, resulting in a consensus model with $1 + m + mn$ nodes. We will assume that each Bottom-Area has the same number of nodes. As a result, the time complexity C can be expressed as follows:

$$C = (m + 1)^2 + (n + 1)^2. \quad (3)$$

B. Experimental

Comparison of the time consumption of the dynamic node joining protocol, firstly compared the performance of VS-PBFT[42] and the improved RSPBFT in this study under different numbers of nodes. fig.4 shows the changes in node joining time consumption of these two algorithms with the change in the number of nodes.

From fig.4, it can be seen that the node joining time consumption of VS-PBFT increases rapidly as the number of nodes increases, while the node joining time consumption of the improved RSPBFT grows more smoothly. This indicates that the improved RSPBFT algorithm proposed in this chapter can better handle the dynamic node joining scenario. Meanwhile, when the number of nodes is relatively small, the node joining time consumption of the two algorithms

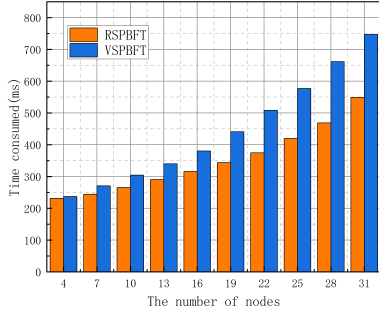


Fig. 4. Node Join's Time Consumed

does not differ greatly. However, as the number of nodes increases, the advantages of the improved RSPBFT algorithm in terms of efficiency and low latency become more apparent. Particularly when the number of nodes is large, the node joining time consumption of the improved RSPBFT algorithm is significantly lower than that of the VS-PBFT algorithm.

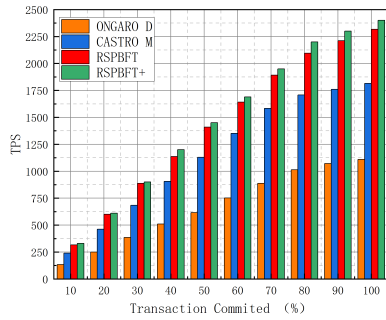


Fig. 5. VRSPBFT's TPS

From fig.5, it can be seen that the addition of the dynamic node joining mechanism has led to better consensus latency performance for VRSPBFT. This is because the addition of the dynamic node joining protocol has solved the issue of consensus restarting due to node joining, which existed in the existing commodity Consortium Blockchain solutions. Meanwhile, under the same transaction submission rate, the improved VRSPBFT algorithm shows better performance in terms of consensus latency and transactions per second.

In summary, the improved RSPBFT algorithm that incorporates the dynamic node joining mechanism and the leader node election protocol has demonstrated significant advantages.

V. CONCLUSION AND FUTURE PROSPECTS

VRSPBFT is an innovative solution aimed at addressing the issues of unreasonable and unfair leader node election, as well as the performance degradation caused by the inability of nodes to dynamically join, which exist in traditional

consortium blockchains. The core feature of this consensus algorithm is the use of dynamic tagging technology to cache security checkpoints, successfully realizing the dynamic joining scenario that cannot be achieved in traditional blockchains, and solving the inefficient behavior of frequent consensus interruptions and restarts due to node entry in the commodity Consortium Blockchain. Furthermore, through the use of VRF, the unfairness issue in leader node election is resolved, thereby maximizing the solution to the "rich getting richer" fairness problem and the single point attack security issue. The experiments have verified the effectiveness of these improvement mechanisms, providing a more robust and efficient consensus solution for consortium blockchains in the complex and dynamic commodity Consortium Blockchain environment. This solution is not only a technological innovation, but also a powerful challenge to the traditional commodity Consortium Blockchain.

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